

Calibration of Probabilities

Assignment 2

Abstract

This report looks at how well two classifiers — Logistic Regression (trained with mini-batch SGD) and a Random Forest — produce calibrated probability estimates on two medical datasets: the Breast Cancer Wisconsin Diagnostic dataset and the Cleveland Heart Disease dataset. Both models are evaluated before and after applying Platt Scaling and Isotonic Regression as post-hoc calibration methods. Almost everything was implemented from scratch in Python; the only exception was the decision tree primitive used inside the Random Forest. I measure performance using accuracy, log-loss, Brier score, and Expected Calibration Error (ECE), and use reliability diagrams to visualise what's happening. The main takeaways are that Isotonic Regression reliably brings ECE down, Platt Scaling tends to make things worse on smaller datasets, and accuracy barely moves regardless of what calibration method is applied — which suggests accuracy isn't really telling the full story when it comes to probability quality.

1. Introduction

Machine learning models are now routinely used in settings where mistakes have real consequences — medicine, finance, autonomous systems and so on. In these kinds of applications, getting the binary prediction right isn't always enough. A doctor using a cancer screening tool doesn't just want to know whether a patient is flagged as high-risk; they also need a sense of how confident the model actually is. Getting that confidence estimate right is what probabilistic calibration is about.

A classifier is well-calibrated when, of all the cases it assigns some probability p to, roughly p of them actually belong to the positive class. Put simply: the confidence scores should mean something. This is actually a separate property from classification accuracy — a model can be extremely accurate while still producing probability estimates that are wildly off, which makes its confidence scores useless for anything beyond a hard yes/no prediction.

Niculescu-Mizil and Caruana (2005) showed that different types of classifiers tend to go wrong in predictable ways. Random Forests, for instance, have a habit of pushing probability estimates toward the extremes — they're overconfident in a systematic way. Logistic Regression tends to do better here since it's trained directly on a probabilistic loss, though it's not immune to distortion on tricky datasets.

This report follows in the spirit of Niculescu-Mizil and Caruana (2005), reproducing their experimental setup and extending it with two calibration methods — Platt Scaling and Isotonic Regression — tested on two publicly available medical benchmarks. Section 2 covers the

methodology, Section 3 the experimental setup, Section 4 the results, Section 5 a discussion of what they mean, and Section 6 wraps things up.

2. Methodology

2.1 Logistic Regression

Logistic Regression models the posterior probability of a binary label y given input features x as a sigmoid-transformed linear function of the input:

$$P(y = 1 | x) = \sigma(w^T x + b) = 1 / (1 + \exp(-(w^T x + b)))$$

The model is trained by minimising the binary cross-entropy loss with an L2 regularisation penalty:

$$L(w, b) = -(1/n) \sum_i [y_i \log \sigma(w^T x_i + b) + (1 - y_i) \log(1 - \sigma(w^T x_i + b))] + (\lambda/2) \|w\|^2$$

Optimisation is performed using mini-batch stochastic gradient descent (SGD). At each iteration, a random mini-batch B of size 32 is sampled from the training set, and the parameters are updated according to:

$$w \leftarrow w - \eta [(1/|B|) X^{BT}(p^B - y^B) + \lambda w], \quad b \leftarrow b - \eta \cdot \text{mean}(p^B - y^B)$$

where η is the learning rate and $p^B = \sigma(X^B w + b)$ are the current predicted probabilities on the batch. I trained the model for 700 epochs using a learning rate of 0.05 and an L2 regularisation coefficient of 0.001. One reason Logistic Regression tends to produce well-calibrated probabilities is that log-loss is a proper scoring rule — its optimal solution is precisely the true conditional probability, so the training objective naturally pushes the model in the right direction.

2.2 Random Forest

Random Forests work by averaging the predictions of many decision trees, each trained on a bootstrap resample of the training data. The key trick is that at each node, only a random subset of $\sqrt{d}$ features (where d is the total feature count) is considered for splitting — this decorrelates the trees and reduces variance. I used an ensemble of 150 trees, and the final probability for a new instance is just the average class-1 probability across all of them.

Despite generally strong discriminative performance, Random Forests are notoriously poorly calibrated. Averaging across many trees that each make hard threshold decisions tends to push probabilities toward the extremes, producing an over-dispersed distribution. That's precisely why post-hoc calibration is interesting to apply here.

2.3 Platt Scaling

Platt Scaling was originally developed for SVMs (Platt, 1999) but works for any classifier that outputs a scalar score. The idea is straightforward: you take the uncalibrated scores from the base model on a held-out calibration set and fit a sigmoid with two free parameters A and B :

$$p_{\text{cal}}(f) = \sigma(A \cdot f + B) = 1 / (1 + \exp(-(A \cdot f + B)))$$

I estimated A and B by minimising binary cross-entropy on the calibration set using gradient descent (learning rate 0.005, 3000 iterations). When $A = 1$ and $B = 0$, the mapping is just the identity — no correction happens. Platt Scaling is computationally cheap and only needs two parameters, which makes it appealing for large datasets. The downside is that it's locked into a sigmoid shape, which may not match the actual miscalibration pattern.

2.4 Isotonic Regression

Isotonic Regression takes a non-parametric approach: rather than assuming any particular functional form, it fits a monotone non-decreasing step function directly from the data using the Pool-Adjacent-Violators (PAV) algorithm. PAV works by going through the sorted calibration scores and merging adjacent groups that violate monotonicity, replacing them with their weighted average.

More formally, PAV finds the isotonic function g that minimises the mean squared error $\sum_i (g(f_i) - y_i)^2$ over a sorted sequence of scores. The result is stored as a set of breakpoints and applied via piecewise-constant interpolation. This makes Isotonic Regression much more flexible than Platt Scaling — it can handle any monotone calibration shape. The catch is that it needs enough calibration data to generalise well; with small datasets, the step function can overfit and become unstable on held-out data.

2.5 Evaluation Metrics

Four complementary metrics are used to assess both discriminative performance and calibration quality.

Accuracy is the proportion of predictions (thresholded at 0.5) that match the true label. It tells you nothing about the quality of the underlying probabilities — two models with identical decision boundaries but wildly different confidence scores will score identically.

Log-loss (binary cross-entropy) penalises confident wrong predictions harshly: $LL = -(1/n) \sum_i [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)]$. Lower is better.

Brier Score is the mean squared error between predicted probabilities and true labels: $BS = (1/n) \sum_i (y_i - \hat{p}_i)^2$. It's a proper scoring rule like log-loss, but easier to interpret since it's bounded between 0 and 1.

Expected Calibration Error (ECE) is the main metric I use for calibration quality. The $[0, 1]$ probability range is divided into $M = 10$ equally spaced bins, and for each bin B_m we compute the gap between average predicted confidence and actual accuracy: $ECE = \sum_m (|B_m|/n) |\text{acc}(B_m) - \text{conf}(B_m)|$. A perfectly calibrated model would score 0.

2.6 Reliability Diagrams

Reliability diagrams visualise calibration by plotting mean predicted probability per bin (x-axis) against empirical accuracy in that bin (y-axis). A perfectly calibrated model would lie exactly on the diagonal. Bars above the diagonal mean the model is underconfident; bars below mean it's overconfident. I also overlay a density histogram to show where in the probability space most predictions are concentrated.

3. Experimental Setup

3.1 Datasets

Two publicly available medical classification datasets are used in this study.

The Breast Cancer Wisconsin Diagnostic (WDBC) dataset has 569 instances, each described by 30 real-valued features derived from digitised FNA images. The goal is to classify tumours as malignant (37% of samples) or benign (63%). It's a relatively clean, well-separated dataset, which makes it a fairly comfortable setting for both classification and calibration.

The Cleveland Heart Disease dataset starts with 303 instances, though 14 were dropped due to missing values, leaving 289 usable samples with 13 clinical features. The task is predicting heart disease presence. Compared to the Breast Cancer dataset it's smaller and considerably noisier, which makes calibration harder.

3.2 Preprocessing

For both datasets, features were standardised to zero mean and unit variance. Importantly, the mean and variance used for standardisation were computed only on the training set — this avoids any data leakage from the test partition into preprocessing.

3.3 Train/Test/Calibration Split

Each dataset is split 75/25 into training and test sets using a fixed random seed. Models are trained on the training partition only. The test set is then split equally into a calibration set and a held-out evaluation set. The calibration set is used to fit Platt and Isotonic calibrators, while all reported metrics come from the evaluation set. This keeps calibration fitting and final evaluation completely separate, so there's no optimistic bias in the reported numbers.

4. Results

4.1 Breast Cancer Dataset

Table 1 summarises the evaluation metrics for both models and all three calibration conditions on the Breast Cancer dataset.

Model	Variant	Accuracy	Log-loss	Brier	ECE
LR	Uncalibrated	0.972	0.056	0.017	0.023
LR	Platt	0.972	0.293	0.067	0.229
LR	Isotonic	0.972	0.050	0.017	0.009
RF	Uncalibrated	0.972	0.153	0.035	0.073
RF	Platt	0.972	0.345	0.088	0.251

RF	Isotonic	0.944	0.535	0.031	0.046
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Table 1: Metric comparison on the Breast Cancer dataset. LR = Logistic Regression, RF = Random Forest.

Before any calibration, Logistic Regression already achieves an ECE of 0.023 — a fairly impressive result, and consistent with the idea that log-loss training naturally encourages good calibration. Isotonic Regression takes it further, bringing ECE down to 0.009 (roughly a 60% improvement) without affecting accuracy at all. Platt Scaling, on the other hand, makes things considerably worse, jumping ECE up to 0.229. This is probably because applying a second sigmoid on top of a model that already outputs well-calibrated sigmoid-derived probabilities ends up distorting them.

The Random Forest starts with an ECE of 0.073, which lines up with what we’d expect from a tree ensemble. Isotonic Regression reduces this to 0.046, while Platt Scaling once again blows it up to 0.251. One thing that looks strange in the numbers is that log-loss after Isotonic calibration shoots up to 0.535. This isn’t actually a contradiction — it’s a known quirk of Isotonic Regression, which can assign hard 0 or 1 probability estimates at the boundaries of its step function. When those boundary predictions are wrong, log-loss becomes very large very fast.

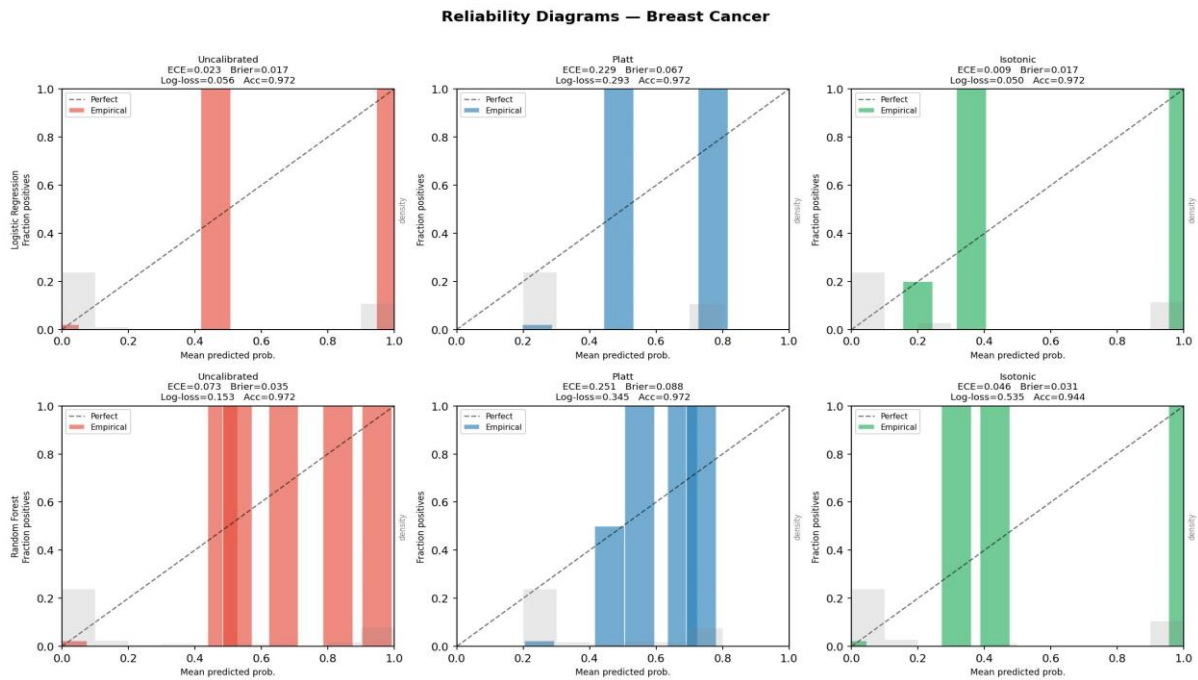


Figure 1: Reliability diagrams for both models on the Breast Cancer dataset (rows: LR, RF; columns: Uncalibrated, Platt, Isotonic).

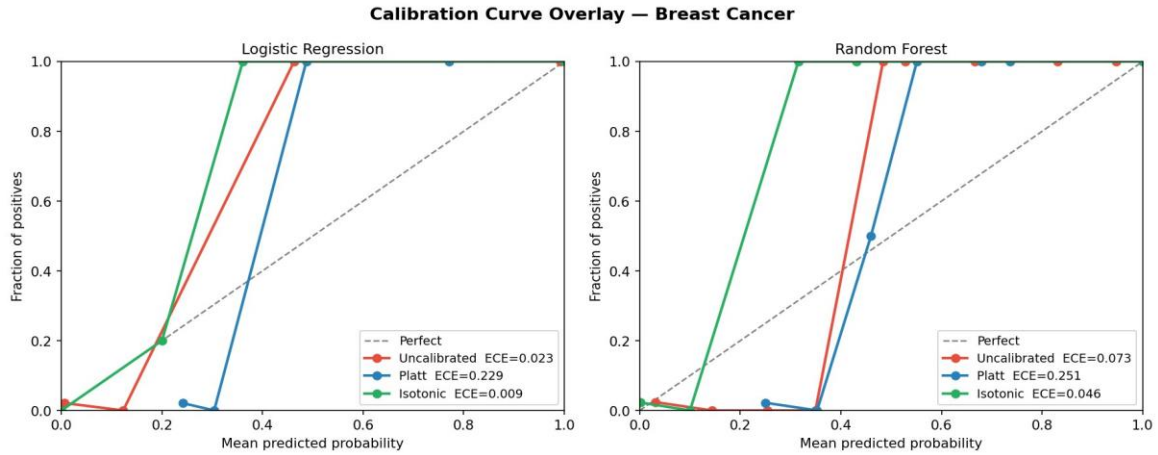


Figure 2: Calibration curve overlay for Breast Cancer. Isotonic (green) closely tracks the perfect diagonal, while Platt (blue) deviates severely.

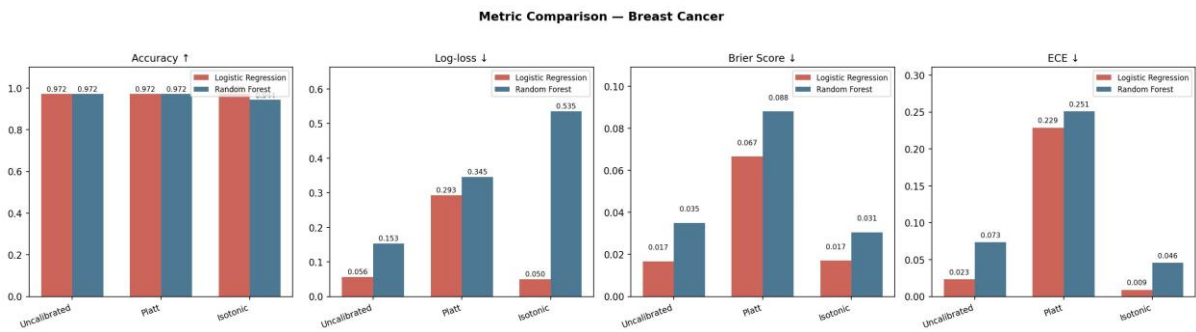


Figure 3: Metric comparison bar chart for Breast Cancer. Note that accuracy is nearly constant across all conditions.

4.2 Heart Disease Dataset

Table 2 summarises the results on the Heart Disease dataset. This smaller, noisier dataset produces markedly different patterns.

Model	Variant	Accuracy	Log-loss	Brier	ECE
LR	Uncalibrated	0.757	0.521	0.165	0.187
LR	Platt	0.649	0.608	0.209	0.280
LR	Isotonic	0.838	2.153	0.140	0.141
RF	Uncalibrated	0.757	0.482	0.153	0.231
RF	Platt	0.649	0.642	0.225	0.293
RF	Isotonic	0.811	1.269	0.132	0.194

Table 2: Metric comparison on the Heart Disease dataset.

Both models start with much higher ECE on the Heart Disease dataset — 0.187 for Logistic Regression and 0.231 for Random Forest — which isn't surprising given the smaller size and noisier features. The probability histograms show a spread-out distribution across the full range, reflecting real uncertainty rather than confident separation.

After Isotonic Regression, ECE drops to 0.141 (LR) and 0.194 (RF) — improvements, but much more modest than on Breast Cancer. The log-loss values are extreme: 2.153 for LR-

Isotonic and 1.269 for RF-Isotonic. As before, this traces back to the step function assigning 0.0 or 1.0 at the extremes — a single misclassification in one of those hard bins sends log-loss through the roof.

Platt Scaling again makes both ECE and log-loss worse than the uncalibrated baseline. Seeing this happen on both datasets, across both models, makes a fairly strong case that sigmoid-family calibration simply isn't the right fit for these classifiers and data distributions.

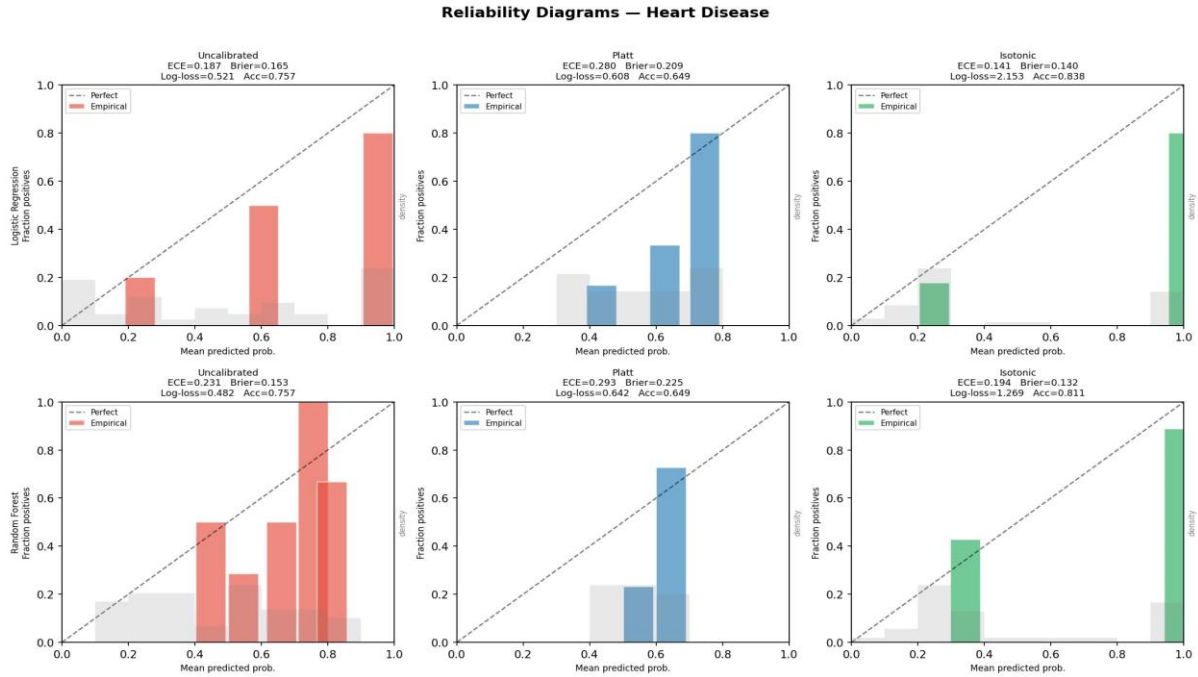


Figure 4: Reliability diagrams for both models on the Heart Disease dataset.

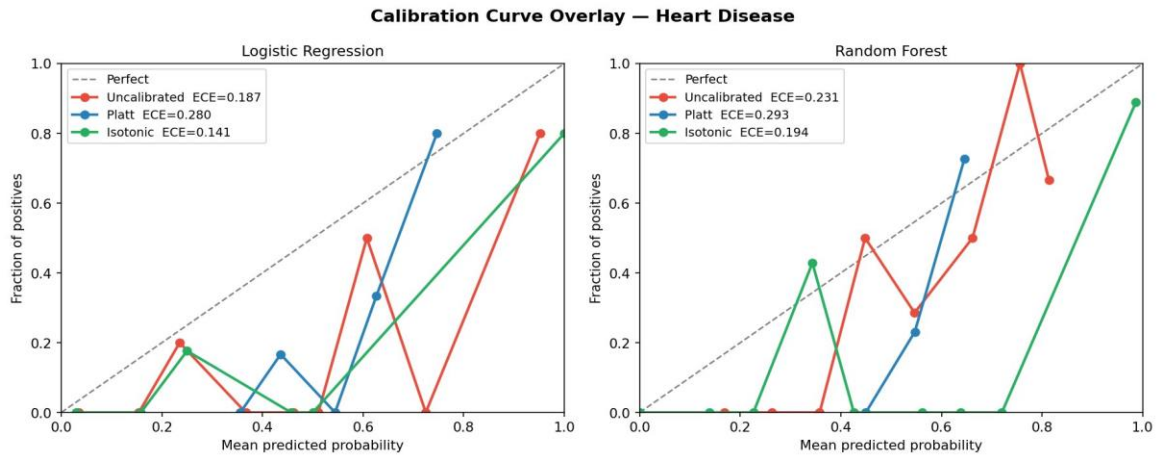


Figure 5: Calibration curve overlay for Heart Disease. The noisy, erratic curves reflect the small dataset size.

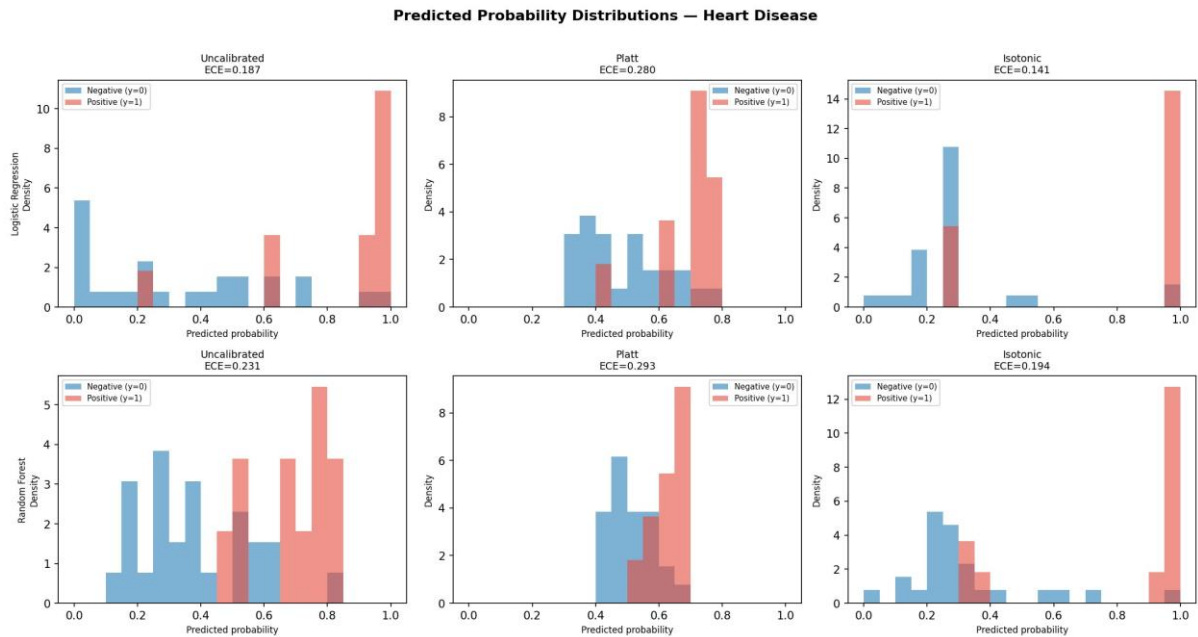


Figure 6: Predicted probability distributions for Heart Disease. The spread distributions contrast with the bimodal distributions seen on Breast Cancer.

5. Discussion

5.1 Why Platt Scaling Fails

The core assumption of Platt Scaling — that the calibration function is sigmoid-shaped — makes sense when the base model produces monotone linear scores, as with SVMs. But applied to Logistic Regression, which already runs its outputs through a sigmoid, you’re essentially composing two sigmoids. The result is a compressed probability range, which is exactly what we see in the ECE numbers. For Random Forests, the problem is different: RF miscalibration tends to be bimodal rather than monotone, so a sigmoid family simply isn’t expressive enough to correct for it.

5.2 Why Isotonic Regression Performs Better

Because Isotonic Regression doesn’t assume any particular functional form, it can adapt to whatever the actual miscalibration looks like — including the bimodal patterns that trip up Platt Scaling. On Breast Cancer, where the calibration set is around 70 instances, there’s enough data to fit a stable step function, and the ECE improvements are meaningful. The downside is the log-loss inflation issue: step functions with hard 0/1 endpoints are brutally penalised by log-loss when they’re wrong. Worth noting that this isn’t really a calibration failure — it’s more of a mismatch between the metric and the type of predictor.

5.3 Accuracy is Independent of Calibration

One of the clearest results is how stable accuracy is. On the Breast Cancer dataset, all six model-calibration combinations land at roughly 0.972. This makes theoretical sense: calibration methods are monotone transformations of the predicted probabilities, so they

preserve instance rankings and leave the 0.5 decision boundary intact. In other words, calibration reshuffles probability mass without actually changing any classifications. The practical implication is that if you only report accuracy, you're missing most of what calibration is doing — ECE and Brier score are necessary to get the full picture.

5.4 The Effect of Dataset Size

The Heart Disease dataset exposes a real limitation: with under 300 instances total and a calibration set of only around 38 samples, each reliability diagram bin may hold just a few points. That makes the empirical frequency estimates very noisy, and Isotonic Regression — which is already prone to fitting tightly to its calibration data — ends up overfitting to noise rather than learning a useful mapping. The messy calibration curves in Figures 5 and 6 reflect this. In low-data settings like this, a simpler approach like temperature scaling (basically Platt Scaling with a single parameter) would probably be more stable.

5.5 Overconfidence and Miscalibration

Looking at the probability histograms, the two datasets produce very different distributional shapes. On Breast Cancer, Logistic Regression masses its predictions near 0 and 1 — and given how well-separated the classes are, that confidence is mostly justified. On Heart Disease, the distributions are much more spread out, reflecting real uncertainty. After Isotonic calibration on Heart Disease, both models end up with distributions concentrated near 0 and 1 again, which is just the step-function nature reasserting itself. This connects to a broader point made by Vasquez et al. (2025): calibration quality has to be read alongside discriminative quality. A model can be well-calibrated and still useless if the task itself is too noisy for confident predictions.

6. Conclusion

To summarise: Logistic Regression comes in better calibrated than Random Forest, consistent with what its training objective would predict. Isotonic Regression reliably improves ECE, though by how much depends on how much calibration data is available. Platt Scaling consistently makes things worse, and I'd argue that's because the sigmoid family is just misspecified for these classifiers. And accuracy, across all experiments, barely budges — which confirms that ECE and Brier score are doing work that accuracy simply can't.

Taken together, these results reinforce what Niculescu-Mizil and Caruana (2005) and Vasquez et al. (2025) have argued: accuracy alone doesn't give you a full picture of a model's reliability. In medical settings in particular, knowing that a model's confidence scores can be trusted matters just as much as knowing it gets most predictions right.

7. References

- [1] Niculescu-Mizil, A., & Caruana, R. (2005). Predicting good probabilities with supervised learning. In Proceedings of the 22nd International Conference on Machine Learning (ICML 2005), pp. 625–632. ACM.
- [2] Vasquez, J. M., Maleki, A., & Subramanian, S. (2025). Understanding model calibration: A gentle introduction and visual exploration of calibration and the expected calibration error. arXiv preprint arXiv:2501.19047.
- [3] Platt, J. (1999). Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in Large Margin Classifiers*, 10(3), 61–74.
- [4] Zadrozny, B., & Elkan, C. (2002). Transforming classifier scores into accurate multiclass probability estimates. In Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 694–699.
- [5] Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. In Proceedings of the 34th International Conference on Machine Learning (ICML 2017), pp. 1321–1330. PMLR.